MariaDB Lab Tasks — Solutions & Implementation Guide

Database Setup, Table Creation, User Privileges, and AI Query Generation

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Task 1: Install MariaDB on Linux and Verify Version

For Debian / Ubuntu systems, update the repository cache, install the MariaDB server package, and ensure the service is enabled and started:

```
sudo apt update
sudo apt install mariadb-server -y
sudo systemctl start mariadb
sudo systemctl enable mariadb
```

For RHEL / CentOS / Fedora distributions:

```
sudo dnf install mariadb-server -y
sudo systemctl start mariadb
sudo systemctl enable mariadb
```

Verify the installation by running the version check command:

```
mariadb --version
```

Task 2: Create Database, Table, and Insert Sample Records

Log into the MariaDB shell as administrative root user:

```
sudo mariadb -u root
```

Execute the SQL script below to create the database foodie_app, create the restaurants table, and populate 3 initial records:

```
-- Create and switch to the database
CREATE DATABASE foodie_app;
USE foodie_app;
```

```
-- Create the restaurants table
CREATE TABLE restaurants (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    cuisine VARCHAR(50) NOT NULL,
    rating FLOAT
);

-- Insert 3 sample restaurant records
INSERT INTO restaurants (name, cuisine, rating) VALUES
('Pasta Fresca', 'Italian', 4.2),
('Spice Symphony', 'Indian', 4.7),
('Taco Fiesta', 'Mexican', 3.8);
```

Task 3: Query Restaurants with Rating Greater Than 4.0

Run a SELECT query filtering restaurants where rating exceeds 4.0 and display only name and cuisine:

```
SELECT name, cuisine
FROM restaurants
WHERE rating > 4.0;
```

Task 4: Grant Permissions to New User and Test

Create the user zomato_user with password zomato123, grant SELECT and INSERT privileges on foodie_app, and apply changes:

```
CREATE USER 'zomato_user'@'localhost' IDENTIFIED BY 'zomato123';
GRANT SELECT, INSERT ON foodie_app.* TO 'zomato_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Connect to MariaDB using the newly created user credentials:

```
mariadb -u zomato_user -p foodie_app
```

Test the granted permissions by inserting a new restaurant and viewing table records:

```
INSERT INTO restaurants (name, cuisine, rating)
VALUES ('Bella Italia', 'Italian', 3.9);
```

```
SELECT * FROM restaurants;
```

Task 5: AI-Assisted SQL Query Generation & Verification

AI Prompt: Prompt provided to ChatGPT / Copilot:

> *"Write a MariaDB SQL query to update the rating of all restaurants with cuisine 'Italian' to 4.5 in the restaurants table."*

Generated UPDATE Query:

```
UPDATE restaurants
SET rating = 4.5
WHERE cuisine = 'Italian';
```

Verification query to confirm the update:

```
SELECT * FROM restaurants WHERE cuisine = 'Italian';
```